MedMNIST Pre-warm-up Exercise: Reproducing 2D and 3D ResNet-50 Benchmarks

Jan Cuesta
Arizona State University

Abstract

This exercise reproduced the MedMNIST v2 ResNet-50 baselines using the authors' released code on the ASU Sol supercomputer. Each of the 12 MedMNIST2D and 6 MedMNIST3D datasets was trained five times for 100 epochs. Test AUC and ACC were summarized as mean and sample standard deviation and compared with the published benchmark. Practical issues involving data paths, Slurm environments, parallel output folders, and ACSConv compatibility were resolved before the final runs. A unified model was also trained across all 18 datasets as a proof of concept for Exercise C.

1 Introduction

MedMNIST v2 contains 12 standardized 2D biomedical image datasets and 6 standardized 3D volume datasets at resolutions of 28×28 and $28 \times 28 \times 28$, respectively [1]. The objective was to reproduce the reported ResNet-50 results using the code released by the authors rather than writing a new training implementation [4].

For each dataset, five independent 100-epoch runs were completed. Test AUC and ACC were recorded, and the mean and sample standard deviation were compared with the published MedMNIST v2 benchmark.

2 Methods

Experiments were executed on ASU Sol using one NVIDIA A100 GPU per job. Long experiments were submitted with `sbatch`, while short smoke tests were used before large submissions.

For 2D datasets, the released script used ResNet-50, image size 28, batch size 128, and 100 epochs. For 3D datasets, the corresponding script used ResNet-50 with `Conv3d`, volume size 28, batch size 32, and 100 epochs. The architecture follows the residual-network design of He et al. [2], while the 3D conversion uses ACSConv [3].

The five final test AUC and ACC values for each dataset were summarized as mean $\pm$ sample standard deviation.

2.1 Problems Encountered

Four main implementation problems were found. First, MedMNIST attempted to download data under `/home`; this was solved by redirecting the MedMNIST root to writable scratch storage. Second, the first Slurm jobs could not import `medmnist`; the batch scripts were changed to call the environment's Python executable directly. Third, simultaneous PathMNIST runs created identical timestamped output folders, causing `FileExistsError`; unique parent output directories were assigned to each run.

Finally, the first 3D smoke test failed inside ACSConv while converting BatchNorm layers under PyTorch 2.14. A minimal compatibility patch was applied so the new Boolean `bias` constructor argument was not populated with the tensor-valued BatchNorm bias. The repeated one-epoch smoke test then completed successfully, after which the full 3D experiments were submitted.

3 Results

Tables 1 and 2 compare the five-run means with the published MedMNIST v2 ResNet-50 baselines [1].

Table 1. MedMNIST2D ResNet-50 results. Current values are mean ± sample standard deviation over five runs.

Dataset	Paper AUC	Current AUC	Paper ACC	Current ACC
PathMNIST	0.990	0.98891 ± 0.00073	0.911	0.90200 ± 0.00942
ChestMNIST	0.769	0.77167 ± 0.00361	0.947	0.94746 ± 0.00039
DermaMNIST	0.913	0.90854 ± 0.00614	0.735	0.73007 ± 0.00541
OCTMNIST	0.952	0.95041 ± 0.00339	0.762	0.76420 ± 0.00923
PneumoniaMNIST	0.948	0.93889 ± 0.01141	0.854	0.84615 ± 0.01116
RetinaMNIST	0.726	0.72659 ± 0.00624	0.528	0.50700 ± 0.03520
BreastMNIST	0.857	0.87149 ± 0.03359	0.812	0.84487 ± 0.02285
BloodMNIST	0.997	0.99745 ± 0.00029	0.956	0.95499 ± 0.00206
TissueMNIST	0.931	0.93079 ± 0.00283	0.680	0.68030 ± 0.00743
OrganAMNIST	0.997	0.99505 ± 0.00071	0.935	0.92342 ± 0.00503
OrganCMNIST	0.992	0.99203 ± 0.00034	0.905	0.90346 ± 0.00544
OrganSMNIST	0.972	0.97209 ± 0.00091	0.770	0.77376 ± 0.00831

Table 2. MedMNIST3D ResNet-50 +3D results. Current values are mean ± sample standard deviation over five runs.

Dataset	Paper AUC	Current AUC	Paper ACC	Current ACC
OrganMNIST3D	0.994	0.99388 ± 0.00070	0.883	0.89574 ± 0.00800
NoduleMNIST3D	0.875	0.88987 ± 0.01501	0.847	0.84903 ± 0.01104
FractureMNIST3D	0.725	0.74456 ± 0.02199	0.494	0.52750 ± 0.05108
AdrenalMNIST3D	0.828	0.85287 ± 0.02067	0.745	0.63423 ± 0.20286
VesselMNIST3D	0.907	0.92168 ± 0.01319	0.918	0.92042 ± 0.02049
SynapseMNIST3D	0.851	0.82391 ± 0.05492	0.795	0.80455 ± 0.02879

Table 3. Final ACC of the unified 100-epoch proof-of-concept.

Dataset	ACC	Dataset	ACC
PathMNIST	0.5195	OrganMNIST3D	0.8541
ChestMNIST	0.9468	NoduleMNIST3D	0.8355
DermaMNIST	0.6863	AdrenalMNIST3D	0.8154
OCTMNIST	0.5830	FractureMNIST3D	0.5000
PneumoniaMNIST	0.8638	VesselMNIST3D	0.9058
RetinaMNIST	0.4975	SynapseMNIST3D	0.7301
BreastMNIST	0.8077		
BloodMNIST	0.7387		
TissueMNIST	0.5077		
OrganAMNIST	0.8235		
OrganCMNIST	0.8649		
OrganSMNIST	0.6890		

4 Exercise C: One Model for All 18 Datasets

What. A unified model was designed with one 2D encoder and one 3D encoder. Both produced a 2048-dimensional feature vector, followed by a shared $2048 \rightarrow 512$ layer and one dataset-specific classification head for each of the 18 datasets.

Why. A single unchanged ResNet-50 cannot directly process both 2D images and 3D volumes, and the datasets also have different numbers and meanings of classes. Separate encoders solve the dimensionality problem, while separate heads solve the different-label-space problem.

How. The dataset identifier selected the correct encoder and output head. Training used 50 sampled batches from every dataset per epoch for 100 epochs, so each task contributed equally to updates of the shared layer. A two-dataset pilot and an all-18 one-batch smoke test confirmed correct routing before the full experiment.

So What. The experiment showed that all 18 datasets can participate in one training framework while preserving modality-specific processing and task-specific outputs. The model was intentionally lightweight, so these ACC values are not directly comparable with the independent ResNet-50 benchmarks. The main limitation is that the dataset ID must be known at inference time. A future version could use larger ResNet-50 encoders or learn an automatic task router.

The 2D reproduction was generally close to the published values. The largest run-to-run variation appeared in smaller datasets such as BreastMNIST and RetinaMNIST. The 3D experiments also reproduced the general benchmark behavior, although AdrenalMNIST3D showed high ACC variability and SynapseMNIST3D showed greater AUC variability.

The final commands did not include every optional preprocessing flag described in the paper, including `--shape_transform` for AdrenalMNIST3D and VesselMNIST3D. Therefore, the results should be interpreted as reproduction of the released-code configuration used in this experiment rather than an exact reconstruction of every paper-specific preprocessing setting.

5 Conclusion

The MedMNIST v2 ResNet-50 exercise was completed with five 100-epoch runs for every 2D and 3D dataset. Most reproduced AUC and ACC values were close to the published benchmark, while the five-run protocol revealed larger variability for a few smaller datasets. The Sol workflow also showed the importance of smoke testing before long GPU jobs. Finally, the all-18 proof-of-concept demonstrated a practical way to combine 2D and 3D tasks using modality-specific encoders, a shared representation, and dataset-specific heads.

References

[1] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2—A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 41, 2023.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, pp. 770–778, 2016.

[3] J. Yang, X. Huang, Y. He, J. Xu, C. Yang, G. Xu, and B. Ni, "Reinventing 2D convolutions for 3D images," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 8, pp. 3009–3018, 2021.

[4] MedMNIST, "MedMNIST Experiments: Training and Evaluation Scripts," GitHub, accessed Sep. 2026.

A Individual Run Results

The following tables list the five individual test results used to compute the mean and sample standard deviation reported in Tables 1 and 2.

Table 4. Individual MedMNIST2D test AUC and ACC values.

	AUC					ACC				
Dataset	R1	R2	R3	R4	R5	R1	R2	R3	R4	R5
PathMNIST	0.98954	0.98980	0.98813	0.98834	0.98875	0.91128	0.89805	0.91072	0.88872	0.90125
ChestMNIST	0.77080	0.76997	0.76802	0.77196	0.77760	0.94791	0.94766	0.94748	0.94686	0.94741
DermaMNIST	0.90795	0.91418	0.90966	0.89845	0.91246	0.72868	0.73566	0.72269	0.73516	0.72818
OCTMNIST	0.95310	0.94704	0.95306	0.94638	0.95246	0.76100	0.76300	0.77300	0.75100	0.77300
PneumoniaMNIST	0.94685	0.93556	0.92090	0.94966	0.94146	0.85737	0.84615	0.84135	0.83013	0.85577
RetinaMNIST	0.72401	0.72989	0.72600	0.71822	0.73482	0.44750	0.53250	0.50250	0.52750	0.52500
BreastMNIST	0.82749	0.84983	0.90100	0.90664	0.87249	0.81410	0.84615	0.84615	0.87821	0.83974
BloodMNIST	0.99775	0.99746	0.99751	0.99696	0.99755	0.95674	0.95528	0.95615	0.95528	0.95148
TissueMNIST	0.92878	0.92828	0.93519	0.93194	0.92978	0.67692	0.67240	0.69211	0.68198	0.67811
OrganAMNIST	0.99531	0.99536	0.99582	0.99477	0.99397	0.92485	0.92766	0.92671	0.92283	0.91506
OrganCMNIST	0.99187	0.99209	0.99153	0.99227	0.99237	0.89387	0.90445	0.90652	0.90689	0.90555
OrganSMNIST	0.97313	0.97196	0.97111	0.97134	0.97292	0.78577	0.76878	0.77138	0.76470	0.77818

Table 5. Individual MedMNIST3D test AUC and ACC values.

	AUC					ACC				
Dataset	R1	R2	R3	R4	R5	R1	R2	R3	R4	R5
OrganMNIST3D	0.99304	0.99388	0.99497	0.99394	0.99358	0.89016	0.89344	0.90984	0.89180	0.89344
NoduleMNIST3D	0.88967	0.89920	0.90377	0.86503	0.89170	0.85161	0.84194	0.83548	0.86452	0.85161
FractureMNIST3D	0.74004	0.72969	0.72237	0.77830	0.75241	0.49167	0.57500	0.52917	0.57917	0.46250
AdrenalMNIST3D	0.82950	0.88343	0.85033	0.86083	0.84026	0.55369	0.77852	0.31208	0.75503	0.77181
VesselMNIST3D	0.91569	0.93202	0.92776	0.93167	0.90128	0.88482	0.92932	0.93455	0.93194	0.92147
SynapseMNIST3D	0.86836	0.84192	0.78636	0.74804	0.87487	0.82955	0.83523	0.79545	0.76420	0.79830